William de Almeida Gilpin

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Education

Stanford University, PhD in Applied Physics, 2019
Stanford University, MS in Applied Physics, 2016
Princeton University, AB in Physics with High Honors, 2014

Positions

The University of Texas at Austin, Assistant Professor of Physics 2022—
Affiliated Faculty, The Oden Institute for Computational Engineering & Sciences.
Harvard University, NSF-Simons Fellow, Quantitative Biology Initiative. 2019–2022.

Visits

Scientific Advisory Boards: Medici Therapeutics (2024—), Radiela Inc (2026–), Causal Labs (2026–)
Max Planck Institute for the Physics of Complex Systems, visiting scientist, spring 2024
Osmosis Medical, content developer, 2018–2020
The University of Tokyo, visiting scholar, fall 2016.
Khan Academy, content developer, 2014 – 2016.

Awards

CNS Teaching Excellence Award
Unconventional AI Grant, 2026
Cottrell Scholar, 2026
NSF CAREER, 2025
Chan-Zuckerberg Investigator, 2023 – 2026
NSF-DMS Digital Twins Grant, 2024
Complex Systems Society Emerging Researcher Award, 2022
Forbes 30 under 30 Science (North America), 2022
APS Prize for Outstanding Doctoral Thesis Research in Biological Physics, 2020.
National Geographic Young Explorers Grant, 2017.
Scientific Visualizations: Nikon Small World; NSF "Vizzies"; APS Milton van Dyke Award (Gallery of Fluid Motion), Physics Today, 2017.
Graduate Fellowships: NSF GRFP, 2014; NDSEG, 2016
Princeton Physics Shenstone & Kusaka Prizes, 2014

Preprints

J. Lai, J. Quinn, A. Bao, **W. Gilpin**. Test-Time Forecasting: Recurrent Depth Enables In-Context Dynamics Prediction. (2026) (arXiv)

J. Lai, A. Bao, J. Quinn, **W. Gilpin**. Fractal basins trap latent reasoning. (2026) (arXiv)

M. S. Krieger, **W. Gilpin**. Discovering causal nonlinear networks from single-cell gene expression. (2026) (bioRxiv)

A. Bao, J. Lai, **W. Gilpin**. Transformers for dynamical systems learn transfer operators in-context. (2026) (arXiv)

J. Aiken, **W. Gilpin**, T. Becker. Expert lithological labels do not improve learning when used to predict marine sediment properties. (2026) (arXiv)

M. S. Krieger, D. Milanova, D. Thompson, N. Cochran, T. Wang, B. Pokharel, R. Ford, A. Gheorghe, **W. Gilpin**, Y. Freund, G. Sugihara, G. Church. Inference and validation of causal genetic interactions in aging human fibroblasts. (2026) (bioRxiv)

Publications

A. Bao, H. Vattikuti, J. Lai, **W. Gilpin**. ***International Conference on Machine Learning (ICML, Spotlight)***. Universal redundancies in time series foundation models. (2026) (arXiv) *Spotlight: highest-rated 2.2% of 23,918 submissions*

J. Lai, A. Bao, **W. Gilpin**. ***International Conference on Learning Representations (ICLR)***. Panda: A pretrained forecast model for chaotic dynamics. (2026) (arXiv) *27% acceptance rate, 19,000 submissions*

Y. Zhang, **W. Gilpin**. ***International Conference on Learning Representations (ICLR)***. Context parroting: A simple but tough-to-beat baseline for foundation models in scientific machine learning. (2026) (arXiv) *27% acceptance rate, 19,000 submissions*

Bauer et al. ***npj Systems Biology & Applications***. Unifying Theories in High-Dimensional Biology. (2026) (arXiv)

W. Gilpin. ***Physical Review X***. Recurrences reveal shared causal drivers of complex time series. (2025) (pdf)

Y. Zhang, **W. Gilpin**. ***International Conference on Learning Representations (ICLR)***. Zero-shot forecasting of chaotic systems. (2025) (pdf) *32% acceptance rate, 11,672 submissions*

W. Gilpin. ***PLOS Computational Biology***. Optimization hardness constrains ecological transients. (2025) (pdf) (cover)

W. Gilpin. ***Bioinformatics***. The cell as a token: high-dimensional geometry in language models and cell embeddings. (2025) (pdf)

Yik et al. ***Nature Communications***. The neurobench framework for benchmarking neuromorphic computing algorithms and systems. (2025) (pdf)

W. Gilpin. ***Nature Reviews Physics***. Generative learning for nonlinear dynamics. (2024) (pdf)

M. J. Falk, F. Roach, **W. Gilpin**, A. Murugan. ***Physical Review Research***. Curiosity-driven search for novel non-equilibrium behaviors. (2024) (pdf)

W. Gilpin. ***Physical Review Research***. Model scale versus domain knowledge in statistical forecasting of chaotic systems. (2023) (pdf)

W. Gilpin. ***Neural Information Processing Systems (NeurIPS)***. Chaos as an interpretable benchmark for forecasting and data-driven modelling. (2021) *21% acceptance rate, 9,122 submissions* (pdf)

W. Gilpin. ***Physical Review Research***. Desynchronization of jammed oscillators by avalanches. (2021) (pdf)

W. Gilpin. ***Neural Information Processing Systems (NeurIPS)***. Deep reconstruction of strange attractors from time series (2020). *20% acceptance rate, 9,467 submissions*. (pdf)

W. Gilpin, Y. Huang, D. Forger. ***Current Opinion in Systems Biology***. Learning dynamics from large biological datasets: Machine learning meets systems biology. (2020) (pdf)

W. Gilpin, M. S. Bull, M. Prakash. ***Nature Reviews Physics***. The multiscale physics of cilia and flagella. (2020) (pdf) (cover)

W. Gilpin. ***Physical Review E***. Cellular automata as convolutional neural networks. (2019) (pdf)

W. Gilpin. ***The Proceedings of the National Academy of Sciences***. Cryptographic hashing using chaotic hydrodynamics. (2018) (pdf)
Press: phys.org, Stanford homepage, KCBS, Futurity, SciShow

W. Gilpin, M. W. Feldman. ***Theoretical Population Biology***. Cryptic selection forces and dynamic heritability in generalized phenotypic evolution. (2018) (pdf)

W. Gilpin, M. W. Feldman. ***PLOS Computational Biology***. A phase transition induces chaos in a predator-prey ecosystem with a dynamic fitness landscape. (2017) (pdf)

W. Gilpin, V. N. Prakash, M. Prakash. ***Journal of Experimental Biology***. Flowtrace: simple visualization of coherent structures in biological fluid flows. (2017) (pdf) (code) (cover)

J. Y. Wakano*, **W. Gilpin*** *(*co-first)*, S. Kadowaki, M. W. Feldman, K. Aoki. ***Theoretical Population Biology***. Ecocultural range-expansion scenarios for the replacement or assimilation of Neanderthals by modern humans. (2017) (pdf)

W. Gilpin, V. N. Prakash, M. Prakash. Rapid behavioral transitions produce chaotic mixing by a planktonic microswimmer. (2018) (arXiv)

W. Gilpin, V. N. Prakash, M. Prakash. ***Nature Physics***. Vortex arrays and ciliary tangles underlie the feeding-swimming tradeoff in starfish larvae. (2017) (pdf)
Press: Nature Physics News & Views, New York Times, Nature, CBS, Popular Science, Business Insider, Scientific American

W. Gilpin, V. N. Prakash, M. Prakash. ***Physical Review Fluids***. Dynamic vortex arrays created by starfish larvae. (2017) (pdf) (aps feature)

W. Gilpin, V. N. Prakash, M. Prakash. ***Nature Physics***. Boundary effects on currents around ciliated larvae. (2017) (pdf)

W. Gilpin, M. W. Feldman, K. Aoki. ***The Proceedings of the National Academy of Sciences***. An eco-cultural model predicts Neanderthal extinction through competition with modern humans. (2016) (pdf)
Press: Newsweek, Science, Daily Mail, Ars Technica, Huffington Post, International Business Times

W. Gilpin. ***Bioinformatics***. PyPDB: A Python API for the Protein Data Bank. (2015) (pdf) (code)

W. Gilpin, S. Uppaluri, C. P. Brangwynne. ***Biophysical Journal***. Worms under pressure: bulk mechanical properties of *C. elegans* are independent of the cuticle. (2015) (pdf) (video)

K. Bayat, W. K. C. Sun, **W. Gilpin**, M. Baroughi, & M. Lončar. ***CLEO: Science and Innovations***. Nitrogen vacancy center ensembles in diamond nanowires. (2014) (pdf)

Patents

W. Gilpin, M. S. Krieger, D. Thompson, D. Milanova, M. Rabinowitz. Methods for inferring nonlinear causal interactions among genes, and related applications thereof. *PCT/US26/47308, filed September 2025*.

Invited Talks

Northwestern University, applied mathematicss colloquium, 2026
Santa Fe Institute, seminar, 2026
NYU Courant, CDS colloquium, 2026
Emory Physics, colloquium, 2026
Berkeley Mechanical Engineering, seminar, 2026
Yale University, Wu Tsai Symposium, 2026
Johns Hopkins Applied Mathematics & Statistics, seminar, 2026
ICTS Bangalore, High-Dimensional Dynamics workshop, 2025
Amazon AWS, 2025
SIAM Applied Dynamical Systems, Minisymposium, 2025
Rice University, biophysics seminar, 2025
NJORD / U. Oslo Conference on Exploring System Dynamics, keynote, 2024
Fields Institute, Symposium on Machine Learning & Dynamical Systems, 2024 (vid)
CZI Theory in Biology Workshop Paros, 2024
Heidelberg University, seminar, 2024
Max Planck Institute for the Physics of Complex Systems, seminar, 2024
UTIG Earthquake Prediction Workshop, 2023
SIAM Applied Dynamical Systems, Minisymposium, 2023
Emory-NSF Multiscale Complex Systems Workshop, 2023
University of Amsterdam, soft matter seminar, 2023
Kungfu AI Inc. machine learning seminar, 2023
U Chicago, Computations in Science seminar, 2022
Pitt-Berkeley-KTH, joint seminar on scientific machine learning, 2022
Ecole Normale Supérieure de Paris, biophysics seminar, 2022
Mila Quebec AI Institute, Université de Montréal, dynamical systems seminar, 2022
Johns Hopkins Physics, seminar, 2021
Flatiron Institute, seminar, 2021

APS March Meeting, Biological physics Dissertation Prize Talk, 2021
UC Santa Cruz Applied Mathematics, seminar, 2021
UT Austin Physics, Center for Nonlinear Dynamics and Biophysics Seminar, 2021
Emory Biology, seminar, 2021
University of British Columbia Mechanical Engineering, seminar, 2021
Brandeis Mathematics, seminar, 2021
Caltech Computing & Mathematical Sciences, CMS Frontiers Colloquium, 2021
UC Berkeley Physics, seminar, 2021
University of Waterloo Applied Mathematics, seminar, 2021
UC Berkeley Chemical and Biomolecular Engineering, seminar, 2021
University of Chicago Physics, seminar, 2021
UC Irvine Mathematics, seminar, 2021
UCLA Mathematics, Applied Math Colloquium, 2021
Princeton University Bioengineering, "Rising Stars" Colloquium, 2020
Microsoft Research New England, 2020
Vrije Universiteit, Stephens group meeting, 2020
MIT Physics, seminar, 2018
Princeton University Physics, PCTS & CPBF Symposium, 2018
Harvard University Quantitative Biology, symposium, 2018
Meiji University, mathematical biology seminar, 2016
Tokyo University of Agriculture and Technology, 2016
National Nanotechnology Infrastructure Network Symposium, 2012
Mote Marine Laboratory, 2011.

Service & Advising

Editorial Advisory Board, *Nature Reviews Physics*.
Research supervisor for four graduate students and three undergraduates. Undergraduates: Yasa Baig (Duke University undergrad); next position: PhD student at Stanford (Marshall Scholar). Glenvelis Perez (Smith College, TACC REU student); next position: graduate student at Stanford Electrical Engineering. Anish Pandya (UT Austin physics undergrad); next position: PhD student at U. Penn. Hasith Vattikuti (UT Austin physics undergrad); next position: PhD student at Caltech.
Search Committee Member. UT Oden Institute / Neuroscience AI Faculty Cluster Search (2024-2025), UT Theoretical Plasma Physics Faculty Search (2022-2023), UT Graduate Admissions (2022-2026).
Grant Referee for U.S. NSF (2025), U.S. DOE (2023), National Geographic Society (2019-2023), European Research Council (2020), Austrian National Academy of Sciences (2024).
Peer review for *Nature Physics*, *The Proceedings of the National Academy of Sciences*, *Nature Communications*, *NeurIPS*, *ICML*, *ICLR*, *Physical Review Letters / X / E / Fluids / Research*, *Nature Reviews Physics*, *PRX Life*, *eLife*, *PLOS Computational Biology*, *Bioinformatics*, *IEEE Transactions on Artificial Intelligence*, *Journal of Experimental Biology*, *Journal of Theoretical Biology*, *Theoretical Population Biology*, *Chaos*, *International Journal of Bifurcation and Chaos*, and *Journal of Archaeological Science*
UT Dissertation Committees: Jake McGrath, Jonas Smucker, Francis Cavanna, James Clarke (Alvarado Lab); Luke Pharr (Marder Group); Erik Hansen (Morrison Group); Yichen Guo (Florin Group); George Yeh (Truskett Group), Edoardo Luna (Thirumalai Group), Jon Staggs (Ostling Group), Simon Kneer (Kantz Group at Max Planck PKS)

Outreach

REU Mentor. Code@TACC Code for Social Change, Oden Moncrief Scholars Program. 2023–2026.
Open source. Sole developer for one open-source tool for bioinformatics, PyPDB, with >150,000 downloads (via PyPI). Group GitHub: (url)
Invited contributor of scientific visualizations: National Geographic Society / Netflix documentary "One Strange Rock"
Educational content developer. Developed educational videos for the non-profit education startups Khan Academy (2014–2016), and Osmosis (2018–2020).
Invited judge for the 2018 American Physical Society "Gallery of Fluid Motion" competition
EDGE-STEM mentor. Mentor and advise early-career doctoral students at Stanford (2015–2019).

Teaching

Computational Physics. Newly-developed & fully open-source graduate course developed in Python, focusing on connections between classical numerical recipes and physical systems. UT Austin, Fall 2022 (17 students), Fall 2023 (28 students), Fall 2024 (32 graduate students and 30 undergraduates). (url)

General Physics for pre-medical majors, single instructor. UT Austin, Fall 2023 (128 students).